



Cambridge (CIE) IGCSE Computer Science



Your notes

Text, Sound & Images

Contents

- * Character Sets
- * Representing Sound
- * Representing Images



Your notes

Character Sets

Character Sets

What is a character set?

- A character set is **all the characters and symbols** that can be **represented** by a computer system
- Each character is given a **unique** binary code
- Character sets are **ordered logically**, the code for 'B' is one more than the code for 'A'
- A character set provides a **standard** for computers to communicate and send/receive information
- Without a character set, one system might interpret 01000001 differently from another
- The number of characters that can be represented is determined by the **number of bits used** by the character set
- Two common character sets are:
 - American Standard Code for Information Interchange (**ASCII**)
 - Universal Character Encoding (**UNICODE**)

ASCII

What is ASCII?

- ASCII is a character set and was an accepted standard for information interchange
- ASCII uses **7 bits**, providing 27 unique codes (128) or a maximum of **128 characters** it can represent



Your notes

Letter	ASCII Code	Binary	Letter	ASCII Code	Binary
a	097	01100001	A	065	01000001
b	098	01100010	B	066	01000010
c	099	01100011	C	067	01000011
d	100	01100100	D	068	01000100
e	101	01100101	E	069	01000101
f	102	01100110	F	070	01000110
g	103	01100111	G	071	01000111
h	104	01101000	H	072	01001000
i	105	01101001	I	073	01001001
j	106	01101010	J	074	01001010
k	107	01101011	K	075	01001011
l	108	01101100	L	076	01001100
m	109	01101101	M	077	01001101
n	110	01101110	N	078	01001110
o	111	01101111	O	079	01001111
p	112	01110000	P	080	01010000
q	113	01110001	Q	081	01010001
r	114	01110010	R	082	01010010
s	115	01110011	S	083	01010011
t	116	01110100	T	084	01010100
u	117	01110101	U	085	01010101
v	118	01110110	V	086	01010110
w	119	01110111	W	087	01010111
x	120	01111000	X	088	01011000

x	120	01111000	X	088	01011000
y	121	01111001	Y	089	01011001
z	122	01111010	Z	090	01011010

Copyright © Save My Exams. All Rights Reserved

- ASCII only represents basic characters needed for **English**, limiting its use for other languages

Extended ASCII

- Extended ASCII uses **8 bits**, providing 256 unique codes ($2^8 = 256$) or a maximum of **256 characters** it can represent
- Extended ASCII provides essential characters such as mathematical operators and more recent symbols such as ©

Limitations of ASCII & extended ASCII

- ASCII has a **limited** number of characters which means it can only represent the **English alphabet**, **numbers** and some special characters
 - A, B, C,, Z
 - a, b, c,, z
 - 0, 1, 2,, 9
 - !, @, #,
- ASCII **cannot** represent characters from languages other than English
- ASCII **does not** include modern symbols or emojis common in today's digital communication

UNICODE

What is UNICODE?

- UNICODE is a **character set** and was created as a solution to the limitations of ASCII
- UNICODE uses a minimum of **16 bits**, providing 2¹⁶ unique codes (65,536) or a minimum of **65,536 characters** it can represent
- UNICODE can represent characters from **all the major languages** around the world



Examiner Tips and Tricks



Your notes

Exam questions often ask you to compare ASCII & UNICODE, for example the number of bits, number of characters and what they store



Your notes

ASCII vs UNICODE

	ASCII	UNICODE
Number of bits	7-bits	16-bits
Number of characters	128 characters	65,536 characters
Uses	Used to represent characters in the English language.	Used to represent characters across the world.
Benefits	It uses a lot less storage space than UNICODE.	It can represent more characters than ASCII. It can support all common characters across the world. It can represent special characters such as emoji's.
Drawbacks	It can only represent 128 characters. It cannot store special characters such as emoji's.	It uses a lot more storage space than ASCII.



Worked Example

The computer stores text using the ASCII character set.

Part of the ASCII character set is shown:

Character	ASCII Denary Code
E	69



Your notes

F	70
G	71
H	72

(a)

Identify the character that will be represented by the ASCII denary code 76 [1]

(b)

Identify a second character set [1]

Answers

(a) L (must be a capital)

(b) UNICODE



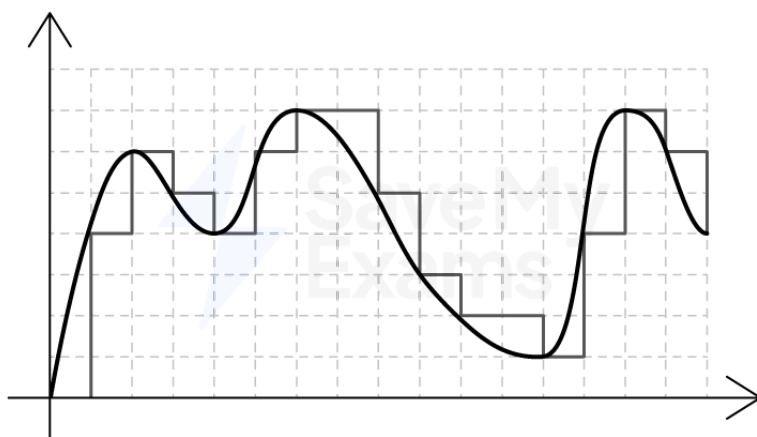
Your notes

Representing Sound

How Sound is Sampled & Stored

How is sound sampled & stored?

- Measurements of the original sound wave are **captured** and **stored as binary** on secondary storage
- Sound waves begin as **analogue** and for a computer system to understand them they must be converted into a **digital** form
- This process is called **Analogue to Digital conversion (A2D)**
- The process begins by **measuring the amplitude** of the **analogue sound wave** at a point in time, called **samples**
- Each measurement (sample) generates a value which can be represented in binary and stored
- Using the samples, a computer is able to create a digital version of the original analogue wave
- The digital wave is stored on secondary storage and can be played back at any time by reversing the process



Copyright © Save My Exams. All Rights Reserved

- In this example, the grey line represents the digital wave that has been created by taking samples of the original analogue wave
- In order for the digital wave to look more like the analogue wave the sample rate and bit depth can be changed

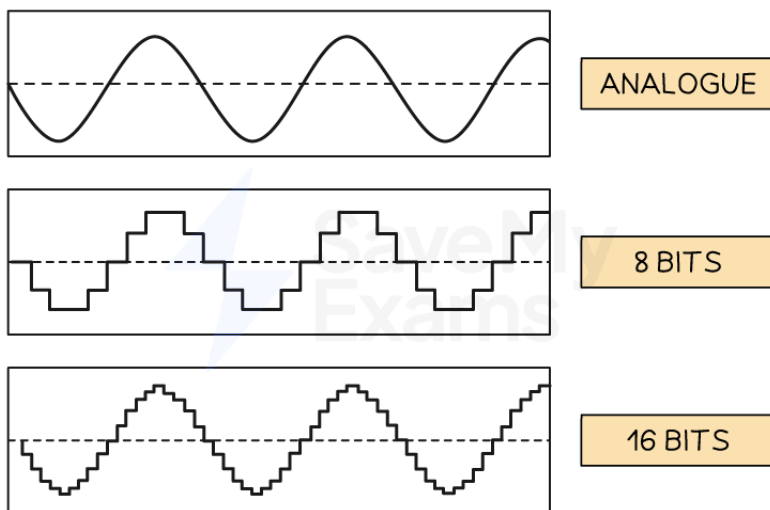


Your notes

Sample Rate & Sample Resolution

What is sample rate?

- Sample rate is the amount of **samples taken per second** of the analogue wave
- Samples are taken each second for the **duration** of the sound
- The sample rate is measured in **Hertz (Hz)**
- 1 Hertz is equal to 1 sample of the sound wave



Copyright © Save My Exams. All Rights Reserved

In this example you can see that the higher the sample rate, the closer to the original sound wave the digital version looks



Your notes



TELEPHONE

16/22 kHz



CD

44.1 kHz



DVD

48 kHz

Copyright © Save My Exams. All Rights Reserved

- The sampling rate of a typical audio CD is 44.1kHz (44,100 Hertz or 44,100 samples per second)
- Using the graphic above helps to answer the question, “Why does telephone hold music sound so bad?”

What is sample resolution?

- Sample resolution is the **number of bits stored per sample** of sound
- Sample resolution is closely related to the **colour depth** of a bitmap image, they measure the same thing in different contexts

What effect do sample rate and sample resolution have?

	Sample rate		Sample resolution	
	High	Low	High	Low
Playback quality	↑	↓	↑	↓
File size	↑	↓	↑	↓



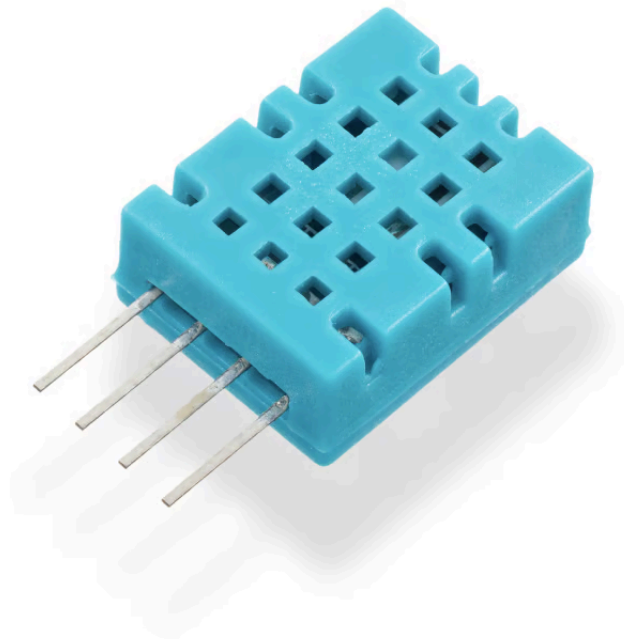
Your notes

Representing Images

Bitmap Images

What is a bitmap?

- A bitmap image is made up of squares called **pixels**
- A pixel is the **smallest element** of a bitmap image
- Each pixel is **stored as a binary code**
- Binary codes are **unique** to the **colour** in each pixel
- A typical example of a bitmap image is a photograph



- The more colours and more detail in the image, the higher the quality of the image and the more binary that needs to be stored

Resolution & Colour Depth

What is resolution?

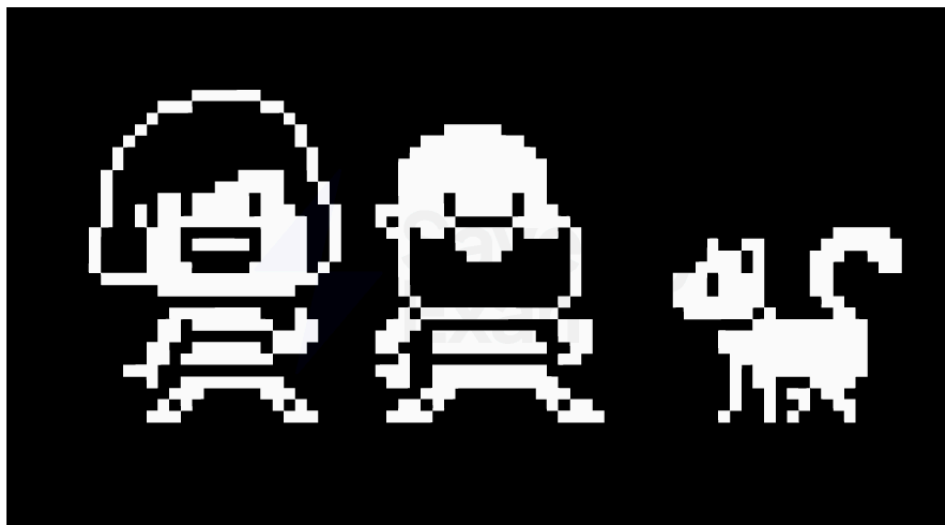


Your notes

- Resolution is the **total amount of pixels** that make up a bitmap image
- The resolution is calculated by **multiplying the height and width** of the image (in pixels)
- In general, the **higher the resolution the more detail in the image** (higher quality)
- Resolution can also refer to the total amount of pixels horizontally in a display, such as:
 - Computer monitors** - 1440p means 1440 pixels horizontally compared to 4K which is 3840 pixels (roughly 4 thousand)
 - TVs** - HD (high definition) channels have a resolution of 1080p, 1080 pixels horizontally compared to newer UHD (ultra high definition) channels with 3840 pixels (4K)
 - YouTube** - The quality button allows a user to change the video playback resolution from 144p (144 pixels horizontally) up to 4K

What is colour depth?

- Colour depth is the **number of bits stored per pixel** in a bitmap image
- The colour depth is dependent on the number of colours needed in the image
- In general, the **higher the colour depth the more detail in the image** (higher quality)
- In a black & white image the colour depth would be 1, meaning 1 bit is enough to create a unique binary code for each colour in the image (1=white, 0=black)



Copyright © Save My Exams. All Rights Reserved

- In an image with a colour depth of 2, you would have 00, 01, 10 & 11 available binary codes, so 4 colours



Your notes



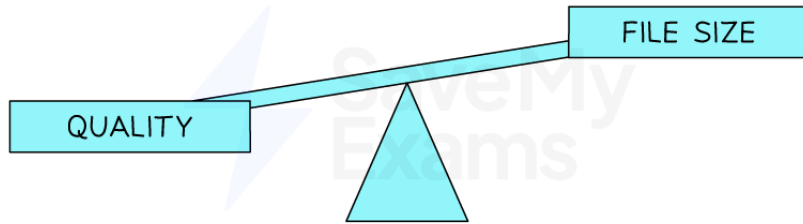
Copyright © Save My Exams. All Rights Reserved

- As colour depth increases, so does the amount of colours available in an image
- The amount of colours can be calculated as 2^n (n = colour depth)

Colour Depth	Amount of Colours
1 bit	2 (B&W)
2 bit	4
4 bit	16
8 bit	256
24 bit	16,777,216 (True Colour)

What is the impact of resolution and colour depth?

- As the **resolution and/or colour depth increases**, the **bigger the size of the file** becomes on secondary storage
- The **higher the resolution**, the **more pixels** are in the image, the more bits are stored
- The **higher the colour depth**, the **more bits per pixel** are stored
- Striking a balance between quality and file size is always a consideration



Copyright © Save My Exams. All Rights Reserved



Your notes



Worked Example

1. Define the term Pixel [1]
2. If an image has a colour depth of 2 bits, how many colours can the image represent? [1]
3. Describe the impact of changing an images resolution from 500x500 to 1000x1000 [2]

Answers

1. The smallest element of a bitmap image (1 square)
2. 4
3. The image quality would be higher [1] the file size would be larger [1]